

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester I Examination 2024-25 (NEP)

DEPARTMENT OF PHYSICS**Paper Code: PHYMJ11 / PHYMN11 — Mechanics & General Properties of Matter**

Time: 2:30 Hours

Maximum Marks: 70

*Instruction: Write your Roll No. at the top immediately on the receipt of this question paper.**Note: All questions carry equal marks (14 marks each). Internal choices are provided.***Question 1** *Attempt any four of the following:* [4 × 3.5 = 14]

- (a) Show that in a perfectly inelastic collision in a laboratory system, there is always a loss of kinetic energy.
- (b) What couple must be applied to a one-meter-long wire, having 1 mm diameter, in order to twist one end of it through 90 degrees, while the other end is fixed? Rigidity of the material of the wire is $\eta = 2.8 \times 10^{10} \text{ N/m}^2$.
- (c) Explain inertial and non-inertial frames of reference with examples.
- (d) A uniform circular ring of mass M and radius R hangs in a vertical plane supported by a knife edge at one point on its inner circumference. Calculate the natural frequency of small oscillation.
- (e) If the centre of mass of three particles of masses 1 kg, 2 kg, and 3 kg is at the point (3, 3, 3), where should a fourth mass of 4 kg be placed so that the centre of mass of the four particles is at the point (1, 1, 1)?
- (f) Given that the orbital velocity of the Sun about the centre of our galaxy is $3 \times 10^7 \text{ cm/s}$ and its distance is nearly equal to $3 \times 10^{22} \text{ cm}$ from the axis of the galaxy, estimate the mass of the galaxy ($G = 6.67 \times 10^{-8} \text{ CGS units}$).

Question 2

- (a) Derive an expression for twisting couple per unit twist for a solid cylinder. Show that for cylinders of the same mass, material and length, twisting couple per unit twist is greater for a hollow cylinder than for a solid one. [7]
- (b) Derive the relation connecting Young's modulus (Y), Bulk modulus (K) and Shear modulus (η) of a material. [7]

Question 3

[14]

Write down the differential equation of a damped harmonic oscillator and solve it. Use the solution to explain three different conditions of damping.

OR

- (a) The frequency of a damped harmonic oscillator is equal to half the frequency experienced by the oscillator without damping. Calculate the logarithmic decrement of the system.
- (b) A harmonic oscillator consisting of a 50 g mass attached to a massless spring has quality factor 200. If it oscillates with an amplitude of 2 cm in resonance with a periodic force of frequency 20 cycles/sec, calculate:
 - (a) the average energy stored in it,
 - (b) the rate of dissipation of energy.

Question 4

- (a) State Bernoulli's theorem. Show that $\frac{1}{2}v^2 + \frac{P}{\rho} + gh = \text{Constant}$, where symbols have their usual meanings. [10]
- (b) A horizontal pipe of a non-uniform bore has water flowing through it such that the velocity of flow is 40 cm/s at a point where the pressure is 2 cm of mercury column. What is the pressure at a point where the velocity of flow is 60 cm/s? (Take $g = 980 \text{ cm/s}^2$ and density of water = 1 g/cm³). [4]

OR

- (a) What are the assumptions of Poiseuille's equation? Write them pointwise. Show that $V = \frac{\pi P r^4}{8\eta l}$. [10]
- (b) Calculate the mass of water flowing in 10 minutes through a tube 0.1 cm in diameter, 40 cm long, if there is a constant pressure head of 20 cm of water. The coefficient of viscosity of water is 0.0089 CGS units. [4]

Question 5

[14]

Obtain an expression for horizontal displacement of an object falling freely under the action of gravity in latitude λ .

Calculate the displacement suffered by a stone dropped from the top of a tower 100 meters high:

- (i) in latitude 60° North,
- (ii) at the equator.
(Angular velocity of Earth about the axis $\omega = 7.2 \times 10^{-5} \text{ rad/s}$ and $g = 9.8 \text{ m/s}^2$).